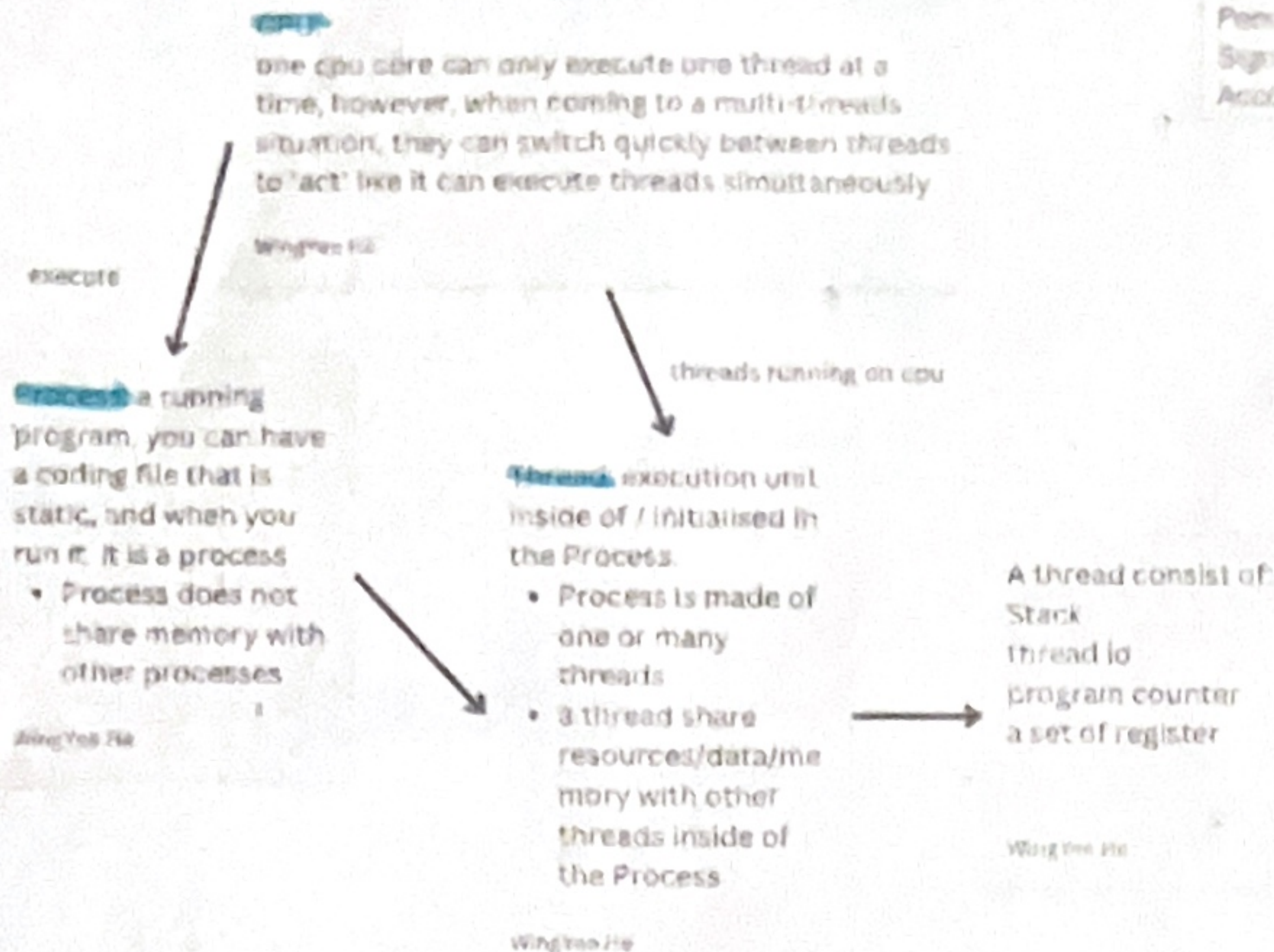


Interrupts are events that cause the CPU not to execute the next instruction. CPU enters Kernel mode and executes an OS interrupt handler.



Address space
Global variables
Open files
Child processes
Pending signals
Signals and signal handlers
Accounting information

Program counter
Registers
Stack
State

NAT

Network Address Translation (NAT) is a networking technique used to modify IP address AND port number information in packet headers while they are in transit across a router or firewall. Its primary purpose is to enable multiple devices on a local network to share a single public IP address when accessing the internet.

网络地址转换 (NAT) 是一种网络技术, 用于在数据包通过路由器或防火墙传输时修改其 IP 地址信息。它的主要目的是让局域网 (如家庭或办公室) 中的多台设备共享一个公网 IP 地址来访问互联网。

Multiple-threads:

- Share an address space.
- Does not provide isolation, e.g., one thread can bring the entire process down

Multiple-processes:

- Enable applications to be parallelised beyond the capacity of a single machine
- Provide isolated execution environments

Thread Control Block (TCB):

stored data include:

- Thread ID, Stack pointer, Program counter, Register values, State (e.g., running, blocked, ready), Pointer to PCB of the process to which the thread belongs to.

Process Control Block

The OS maintains information about a process in a data structure called the Process Control Block (PCB).

data stored in the PCB include:

- Process ID, process state, parent process, memory management information, file descriptors, priority, used CPU time, etc.

Race condition: Occur when multiple processes/threads access a shared resource. The results depend on the timing of the execution, the order in which instructions are executed

Priority Inversion Problem: Process with higher priority always get to run as long as it is not blocked

Non-preemptive:

Pick a process to run and let it run until it blocks or voluntarily releases the CPU. Context switches are performed only when necessary

- First-come First-served
- Shortest Job First

Preemptive:

- Pick a process and let it run for a maximum of some fixed time (a quantum). Suspend it when reached quantum, and picks another process to run. Clock interrupts are needed

- Round Robin
- Priority Scheduling
- Multi-level Feedback Queue: It has multiple queues, each with a different priority. The lower the priority, the higher the quantum is given; Balances responsiveness and throughput

Turnaround time = time complete - time arrived

Strict Alternation

Busy waiting approach in which 2 processes/threads take turns running the critical region.

Thread A gets to enter the critical region when turn is 0:

```
int turn; // global variable
...
void threadA()
{
    while(TRUE) {
        while (turn != 0) {}
        criticalRegionA();
        turn = 1;
        nonCriticalRegionA();
    }
}
```

Thread B gets to enter the critical region when turn is 1:

```
int turn; // global variable
...
void threadB()
{
    while(TRUE) {
        while (turn != 1) {}
        criticalRegionB();
        turn = 0;
        nonCriticalRegionB();
    }
}
```

Yield Mutex

Blocking approach in which When a thread wants to enter its critical region. It checks if entry is allowed. If not, let another thread use the CPU while waiting for the lock.

Busy waiting: if entry not allow, executes a loop, continuously checking process

Strict Alternation: Busy waiting approach

Mutual Exclusion.

- No 2 processes in Critical Region at the same time.
- No assumption about speeds / num of CPUs
- No process running outside C.R. should block others
- No process should have to wait forever to enter C.R.

Physical address = (frame number x frame size) + offset

Translation Lookaside Buffer (TLB) is a small, high-speed cache in a computer's CPU that helps speed up memory access by storing recently used translations of virtual addresses to physical addresses.

Page Replacement Algorithm - **Second-chance**

Algorithm

Simple modification to FIFO to avoid evicting a recently used page

- Inspect the Referenced (R) bit of the oldest page
 - If 0, the page is old and not recently used, evict
 - If 1, the page is old but recently used
- Clear the R bit
- Move the page to the tail of the queue (give it a second chance)
- Update the load time of the page as if it had just arrived in memory

Symmetric encryption (use AES algorithm)

- Uses the same key for encryption and decryption
- The key must remain secret

Asymmetric encryption (use RSA algorithm)

- Uses two different keys
- A public key to encrypt and a private key to decrypt
- The private key must remain secret

Symmetric encryption is efficient for encrypting large messages but requires securely exchanging a secret key; **Asymmetric encryption** does not require securely exchanging a secret key but is not suitable for encrypting large messages.

Process - key exchange Protocol - using Asymmetric Encryption

- Alice and Bob want to exchange messages using symmetric key encryption
 1. Alice generates a key pair ($K_{public}^A, K_{private}^A$)
 2. Alice posts her public key online
 3. Bob obtains K_{public}^A
 4. Bob generates a secret key K_{secret}
 5. Bob computes the ciphertext of the secret key $c = \text{encrypt}(K_{secret}, K_{public}^A)$
 6. Bob sends c to Alice
 7. Alice recovers K_{secret} by computing $\text{decrypt}(c, K_{private}^A)$
 8. Alice and Bob share a secret key and can exchange messages using symmetric encryption

Message Authentication Code (MAC) Used to detect if a message has been tampered with. MAC function generate a tag t given a message and a secret key

Digital Signatures

Asymmetric cryptography: - A private signing key - A public verification key

之前, public key is for encrypt and private key is for decrypt, but in here, private key is for sign, and public key is for verification

Advanced Encryption Standard - AES

two types: ECB (Electronic Codebook) & CBC (Cipher Block Chaining)

ECB ✗

Each block is encrypted and decrypted independently using K_s

Problem: two same plain text will encrypted into the two same cipher text. We are leaking information.

CBC ✓

XOR Operation
 $1 \text{ XOR } 1 = 0$ $0 \text{ XOR } 0 = 0$
 $0 \text{ XOR } 1 = 1$ $1 \text{ XOR } 0 = 1$
Based on probabilistic encryption $1 \text{ XOR } 1 = 0$

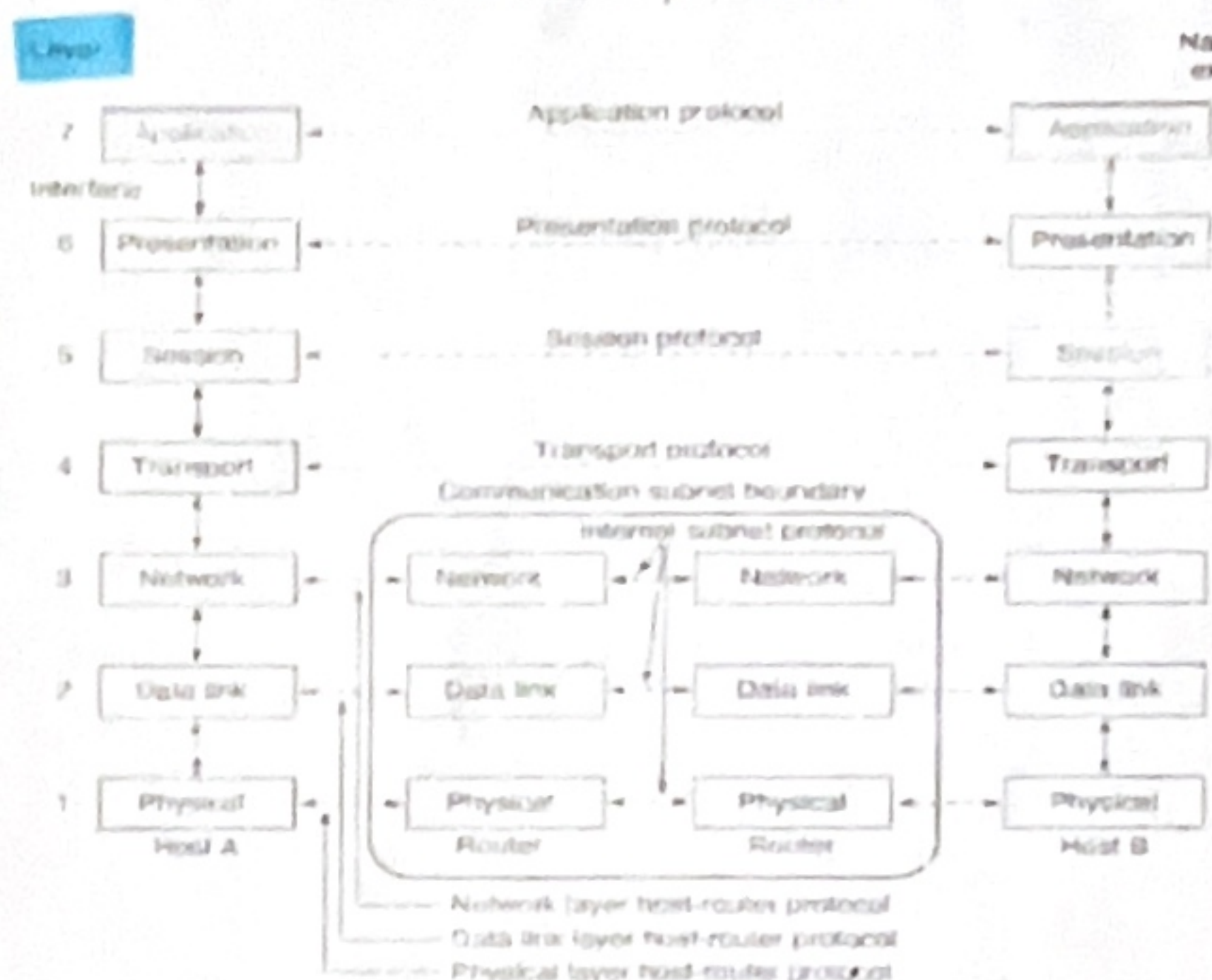
- The use of randomness during encryption so that encrypting the same plaintext results in different cyphertexts

- Encryption must be done sequentially
- Decryption can be done in parallel

三大Principles

- **Confidentiality**: Only the sender and intended receiver should be able to understand the contents of the communication
- **Integrity**: Detect if the content of the communication has been tampered with
- **Authentication**: Establish the identities of one or both of the endpoints to confirm that they are who they claim to be

A socket has **5 tuple**: Protocol, local/remote IP, local/remote port number



Simple mail transfer protocol (SMTP)

SMTP uses TCP to reliably transfer email message from client to server. Transfer involves

- (1) handshaking
- (2) transfer of messages
- (3) closure

Internet message access protocol (IMAP)

IMAP enables users to interact with Mail Transfer Agent, a server. Essentially, IMAP provides a way for users to view and manipulate their emails stored remotely on the mail server.

multiplexing/demultiplexing is about get data from computer to another computer's application.

UDP - User Datagram Protocol:

Carries conversation without connection

UDP sends data by using UDP header (contains ^{source} source and destination ports) follow by Payload (the core data that we are sending)

TCP - Transmission Control Protocol Features:

Full duplex - data in both directions simultaneously

Has exact pairs of senders and receivers

Bytes streams, not message streams

Flooding: A simple adaptive routing that guarantees shortest distance. Robust, but inefficient. basically an exhaustive solution. Have to have a way of discarding packets (a Time To Live field in each packet)

Dijkstra's Algorithm

Divide nodes into three groups: "unseen", "open", "closed"

- unseen: Not a neighbour of any node we have processed
- Open: We have "visited" a neighbour, but not it. We know a path
- Closed: We have visited it. We know the best path to it

The algorithm moves nodes: unseen -> open -> closed

initially no paths are known so all nodes have a value of infinity ("unseen")

As the algorithm proceeds, labels will be updated

- Tentative ("open") - initial state
- Permanent ("closed") - once a shortest path is found label is permanent and won't change again

Link State Routing

- Simple 5-step process at each router;

1. Discover its neighbours and learn their network address
2. Set the distance or cost metric to each of its neighbours
3. Construct a packet containing all it has just learned
4. Send the packet to, and receive packets from, all other routers
5. Compute the shortest path to every other router

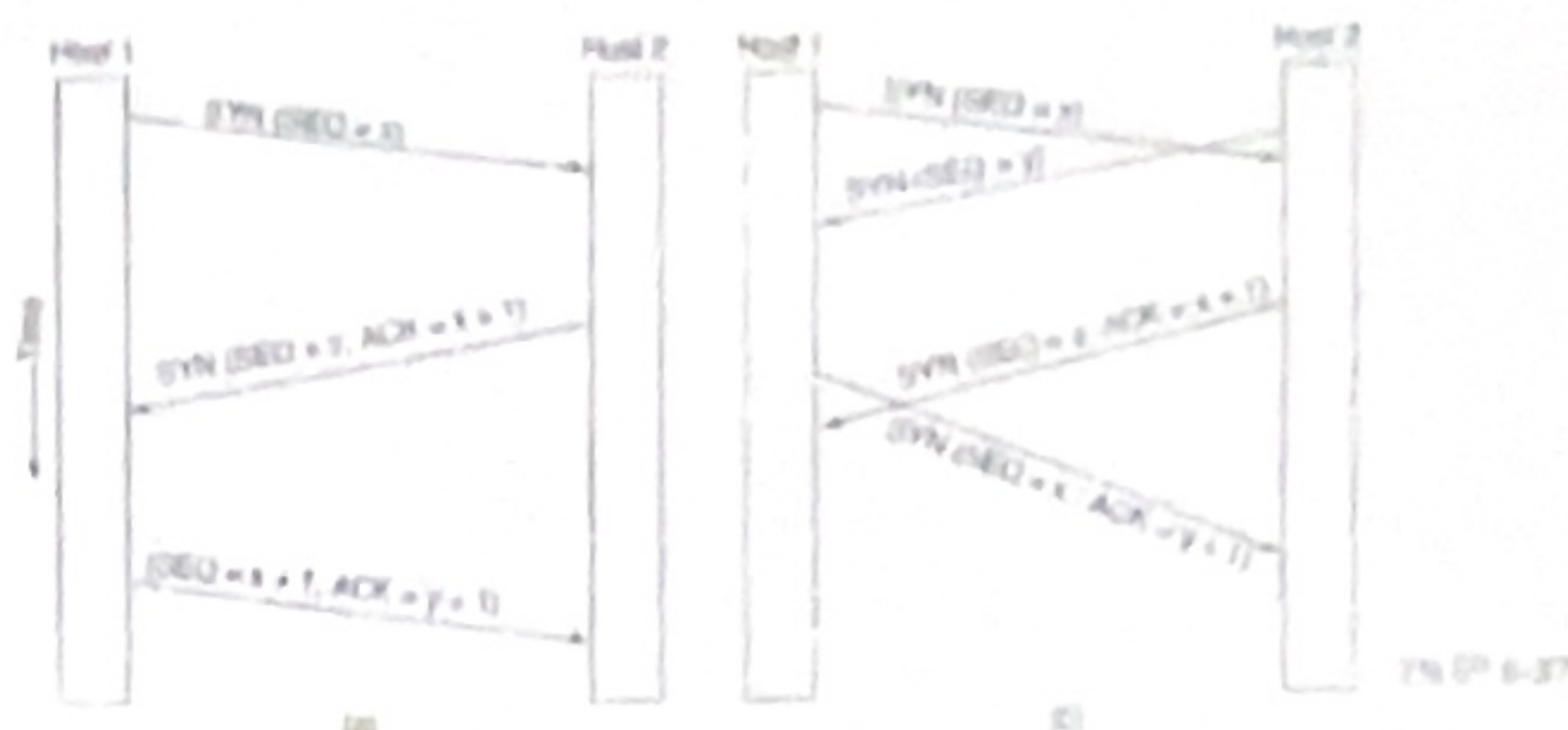
A routing protocol.
each node learns entire topology of the network and contain shortest path from itself to all other nodes

Border Gateway Protocol (BGP)

A protocol for external routing between Autonomous Systems

- In contrast to the internal routing, BGP needs to consider politics as well
- Companies not willing to have their network used by others.
- Border Gateway Protocol typically based on
 - customer/provider: I pay you for "transit" of traffic I send or receive
 - or peering agreements: we carry each others' traffic without charge

Three-way handshake: ensure only one connection is establish, and establish initial sequence numbers for sliding window



Normal operation

Simultaneous connection attempts.

Acknowledge number: use to indicate willing to accept connection, and represents the next byte the sender expects to receive

Datagram Network Virtual Circuit

Host	Datagram Network	Virtual Circuit
Type	Connectionless	Connection-oriented
State	Router do not hold state information about connections	Each VC requires router table space Router detects a problem.
Addressing	(-) Each packet has full source and destination	(+) Each packet contains a short VC number
Routing	Each packet independently	Defined at set up
Quality of Service	- Difficult	+ Easy if enough resources
Congestion control	- Difficult	+ Easy if enough resources
Link failure recover	+ Simple	+ Extra work

Connectionless Network - datagram network

Connection-oriented network - Virtual Circuit (VC) network

Internet Control Protocols

- **ICMP Internet Control Message Protocol:** used for sending error messages and operational information. For example, if a router can't deliver a packet, uses ICMP to send a message back to the sender.
- **DHCP Dynamic Host Configuration Protocol:** DHCP automatically assigns IP addresses to devices (such as computers, printers, smartphones) when they join a network (like uni wifi).
- **ARP Address Resolution Protocol:** It translate an IP address into a MAC address. MAC (Media Access Control) address is assigned by the hardware manufacturer of your device's network interface card (NIC)

